ABHIRAM PATIL

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EDUCATION

Bachelor of Computer Science, MIT World Peace University — 2023 – Present
Pune, India — CGPA: 7.49

SKILLS

CORE SKILLS	Python, FastAPI, REST APIs,Backend Development
AI & Data	RAG pipelines, LLM integration, NLP, Google STT/TTS, Vertex AI
Real-Time Systems	LiveKit Agents, WebRTC, audio streaming pipelines
Databases	SQL, PostgreSQL
Tools	Git/GitHub, Docker, Postman, Render, Vercel

PROJECTS

Real-Time AI Voice Agent (Backend + AI) — GitHub

Python, FastAPI, LiveKit, WebRTC, Google Cloud STT, Vertex AI Gemini, Google Cloud TTS,

- Built a real-time voice pipeline using LiveKit, streaming audio through STT → LLM → TTS with end-to-end latency under 3 seconds.
- Integrated Google Cloud STT, Vertex AI LLM, and TTS to enable low-latency, fully voice-driven conversational interactions.
- Implemented JWT-based authentication for WebRTC sessions and configured secure LiveKit room access.
- Designed event-driven backend logic handling audio streams, transcription events, and LLM responses.

Retrieval-Augmented Generation (RAG) System — GitHub

Node.js, Express.js, Python, Flask, PostgreSQL, pgvector, Ollama,Docker

- Developed a multi-service backend system integrating a Node.js/Express backend, a Python AI microservice, and PostgreSQL with pgvector to enable Retrieval-Augmented Generation (RAG) and semantic search.
- Engineered a multi-service request workflow where the Express backend handled client requests and data persistence before delegating AI processing to a dedicated Python microservice, isolating AI responsibilities from the core application logic.
- Developed an AI ingestion pipeline that transformed project data into vector embeddings using Ollama and indexed them in PostgreSQL with pgvector for efficient semantic retrieval.
- Implemented semantic search and Retrieval-Augmented Generation (RAG) by retrieving relevant project context to generate grounded, context-aware AI responses.
- Designed an asynchronous AI ingestion workflow that decoupled CRUD operations from embedding generation, preventing AI processing from blocking core application workflows

Sentiment & Emotion Analysis (Twitter NLP Tool) — GitHub

Python, Tweepy, Scikit-learn, Pandas, NLTK, Matplotlib

- Built a Python-based sentiment analyzer fetching and processing 1,000+ tweets using the Twitter API.
- Implemented an NLP + supervised ML pipeline that achieved 90%+ accuracy on labeled sentiment data.

ACHIEVEMENTS

- Participant in national hackathons including Smart India Hackathon (SIH) and HACKMIT.
- Certified in Data Visualization with Excel and Cognos (IBM), and in Critical Thinking & Teamwork (Rochester Institute of Technology).